Yeonju Lee

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RESEARCH INTERESTS

Knowledge-Informed Machine Learning for Complex Systems

Developing learning methods that integrate domain and scientific knowledge to improve reliability under limited data and changing environments.

- Domain-guided learning from imaging and spatiotemporal sensor data
- Foundation models for knowledge-grounded scientific reasoning
- Robust learning-enabled decision-making and control in physical systems

Applications: Healthcare and Precision Agriculture

EDUCATION

Georgia Institute of Technology (Expected) 2027
Ph.D. in Machine Learning, Advisor: Dr. Jing Li
(Home Department: ISyE)

Yonsei University 2022
M.S. in Industrial Engineering

Yonsei University 2021
B.S. in Information and Industrial Engineering

RESEARCH PAPERS

(P1) **Y. Lee**, X. Qiao, Y. Shi, W. Liang, Y. Chen, J. Li, LLM-Guided Knowledge Alignment for Robust Learned Control in Complex Physical Systems. *Manuscript in preparation.*

(P2) **Y. Lee**, X. Qiao, Y. Shi, W. Liang, Y. Chen, J. Li, Depth-Specific Soil Moisture Trajectories for Cross-Year Sugar Beet Yield Prediction. *Manuscript in preparation.*

(P3) **Y. Lee**, R.Q. Chen, X. Qiao, Y. Shi, W. Liang, Y. Chen, J. Li, SPADE: A Large Language Model Framework for Soil Moisture Pattern Recognition and Anomaly Detection in Precision Agriculture. Revised manuscript under review in *Computers and Electronics in Agriculture*, 2026.

(P4) **Y. Lee**, R.Q. Chen, H. Yan, M. Mupparapu, F. Lure, J. Li, F.C. Setzer, Dentin-Normalized CBCT Radiomics within a Periapical Lesion Differential Diagnosis System for Differentiating Granulomas and Radicular Cysts. *Journal of Endodontics*, 2026.

(P5) M.G. Kwak, **Y. Lee**, H. Wang, J. Li, BrainNormalizer: Anatomy-Informed Pseudo-Healthy Brain Reconstruction from Tumor MRI via Edge-Guided ControlNet. *IISE Transactions on Healthcare Systems Engineering*, 1–13, 2026.

- *Selected for the journal's Monthly Spotlight Series*, 2026

(P6) R.Q. Chen, **Y. Lee**, J. Li, The Science Behind Machine Learning, Deep Learning, and Active Learning. *Dental Clinics of North America*, 70(2), 351–359, 2026.

(P7) A. Cohen, Y. Sun, Z. Qin, H. Muriki, Z. Xiao, **Y. Lee**, M. Housley, A.F. Sharkey, R.S. Ferrarezi, J. Li, G. Lu, Y. Chen, Modular, On-Site Solutions with Lightweight Anomaly Detection for Sustainable Nutrient Management in Agriculture. *ACS ES&T Engineering*, 6(3), 1089–1105, 2026.

(P8) R.Q. Chen, **Y. Lee**, B. Joffe, P.C. Costa, C. Filan, B. Wang, S. Balakirsky, F. Robles, K. Roy, J. Li, Foundation model cascades enable zero-shot microscopy image analysis for cell therapy manufacturing. *Cytotherapy*, 28(5), 102078, 2026.

(P9) **Y. Lee**, M.G. Kwak, R.Q. Chen, H. Yan, M. Mupparapu, F. Lure, F.C. Setzer, J. Li, Oral-Anatomical Knowledge-Informed Semi-Supervised Learning for 3D Dental CBCT Segmentation and Lesion Detection. *IEEE Transactions on Automation Science and Engineering*, 22, 11205–11218, 2025.
- *IISE DAIS Best Paper Finalist*, 2024

(P10) R.Q. Chen, **Y. Lee**, H. Yan, M. Mupparapu, F. Lure, J. Li, F.C. Setzer, Leveraging Pretrained Transformers for Efficient Segmentation and Lesion Detection in Cone-Beam Computed Tomography Scans. *Journal of Endodontics*, 50(10), 1505–1514, 2024.

(P11) **Y. Lee**, Y. Kim, B. Lee, C.O. Kim, Discovery of Fault-Introducing Tool Groups with a Numerical Association Rule Mining Method in a Printed Circuit Board Production Line. *International Journal of Production Research*, 62(9), 3305–3319, 2024.

PATENTS AND PATENT APPLICATIONS

- R.Q. Chen, J. Cheng, **Y. Lee**, J. Li, F. Lure, M. Mupparapu, F.C. Setzer, and H. Yan, "Leveraging Pre-Trained Transformers for Efficient Segmentation and Lesion Detection in Cone-Beam CT Images." U.S. Provisional Patent Application No. 63/654,892, filed May 31, 2024.
- **Y. Lee**, J. Cheng, M.G. Kwak, R.Q. Chen, J. Li, F. Lure, M. Mupparapu, F.C. Setzer, and H. Yan, "Oral-Anatomical Knowledge-Driven Semi-Supervised Segmentation for 3D Dental CBCT Images." U.S. Provisional Patent Application No. 63/654,901, filed May 31, 2024.

HONORS & SELECTED AWARDS

- **George Fellowship**, H. Milton Stewart School of Industrial & Systems Engineering, 2025
- **Stewart Fellowship**, H. Milton Stewart School of Industrial & Systems Engineering, 2022, 2025
- **Finalist for Best Track Paper Award**, IISE DAIS, 2024
- **Accelerated Degree Program Scholarship**, Yonsei University, 2021

CONFERENCE TALKS

(C1) Knowledge-Informed Deep Learning for Periapical Lesion Diagnosis in Cone Beam Computed Tomography Scans. *INFORMS Annual Meeting*, Oct 2025, Atlanta, USA (invited).

(C2) Sugar Beet Yield Prediction Using Partial Least Square Regression on Root Zone Soil Moisture Patterns. *Quad-AI ENGAGE Workshop on AI and Digital Agriculture*, Jun 2025, Atlanta, USA.

(C3) Comparison of automated versus clinician-based outcome assessment after endodontic surgery. *Penn Dental Medicine Advances in Clinical Care and Education (ACCE) Day*, May 2025, Pennsylvania, USA.

(C4) Deep Learning for Three-Dimensional Semantic Segmentation for Periapical Lesion Detection on Cone-Beam Computed Tomography. *SIIM Annual Meeting*, Jun 2024, Maryland, USA.

(C5) Oral-Anatomical Knowledge-Informed Semi-Supervised Learning for 3D Dental CBCT Segmentation and Lesion Detection. *IISE Annual Conference*, May 2024, Montréal, Canada.

TEACHING

- **Guest Lecturer** at Georgia Tech — Spring 2025
  Basic Statistics Method (ISYE 3030)
  Delivered three lectures on statistical foundations and advanced machine learning (Teaching Evaluation: 4.8/5.0)
- **Tutor** at Georgia Tech — Fall 2022, Spring 2023
  Regression and Forecasting (ISYE 4031)
- **Teaching Assistant** at Yonsei University — Spring 2022
  Machine Learning 2 (DSS 6017)
- **Guest Lecturer** in SK Hynix-sponsored professional training program — Winter 2021
  Deep Learning with Python
- **Teaching Assistant** at Yonsei University — Fall 2021
  Production Process Control (IIE 3104)

GRANT PROPOSAL EXPERIENCE

NIH SBIR: Integrated AI Prognostication Framework for Endodontic Treatment Outcome
Literature review and research gap writing, dental AI prognostic framework development, and preliminary predictive modeling.

NSF TTP: DeepYield-TR: An AI-driven sucrose prediction, practice benchmarking, and decision support system for sugar beet growers
Literature review and technical writing on research translation.

PROFESSIONAL SERVICE

- Reviewer: IEEE TASE, IEEE TMI
- Member: INFORMS, IISE

Last updated Aug 2026.